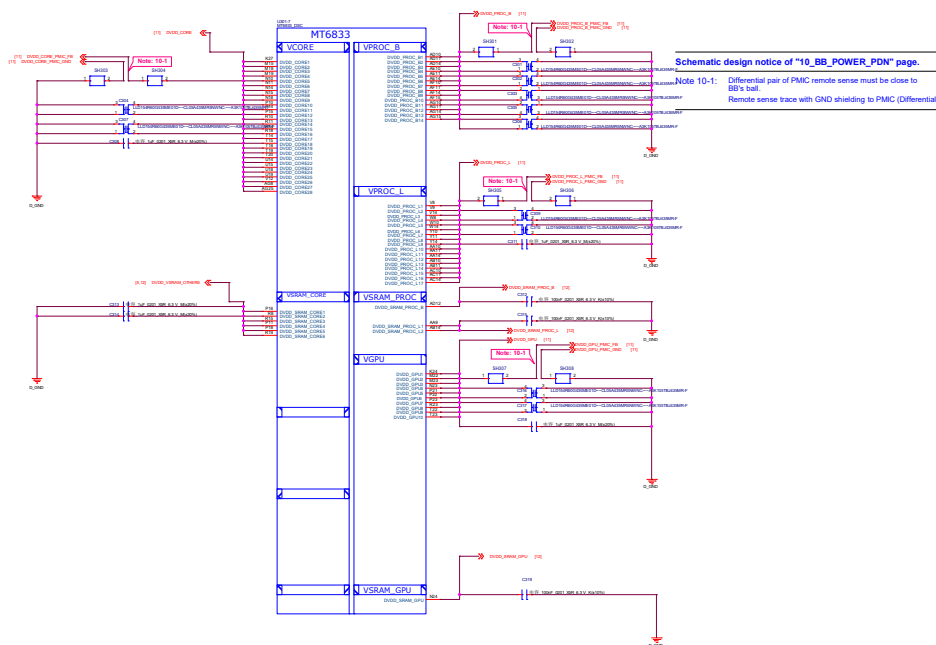
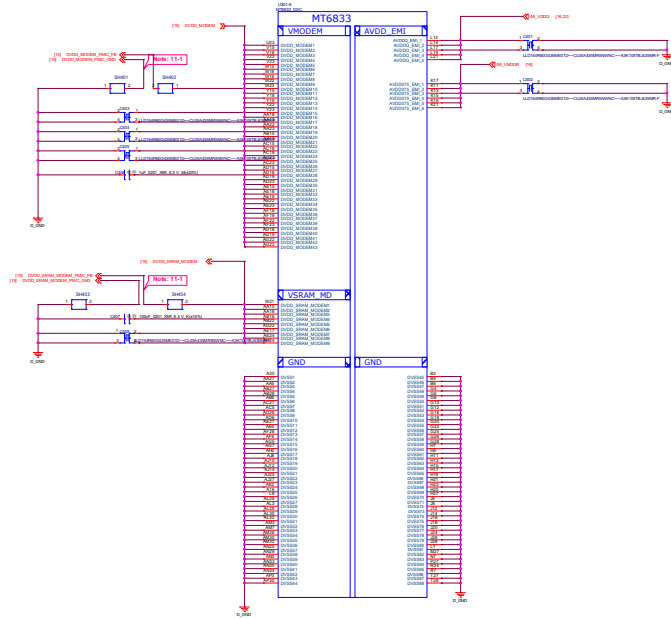


Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1	Disconnect Dummy pins A1,A34,AP1,AP34 to GND
		29_POWER_EXT	Change C2923 cap from 6.3V to 10V
		20_POWER_MT6365_Buck	[PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC
		25_POWER_MT6315_VMD	[PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title SM6350_Control		
Size D	Project <Doc>	Rev V1
Date: Monday, October 18, 2021	Sheet 2 of	50

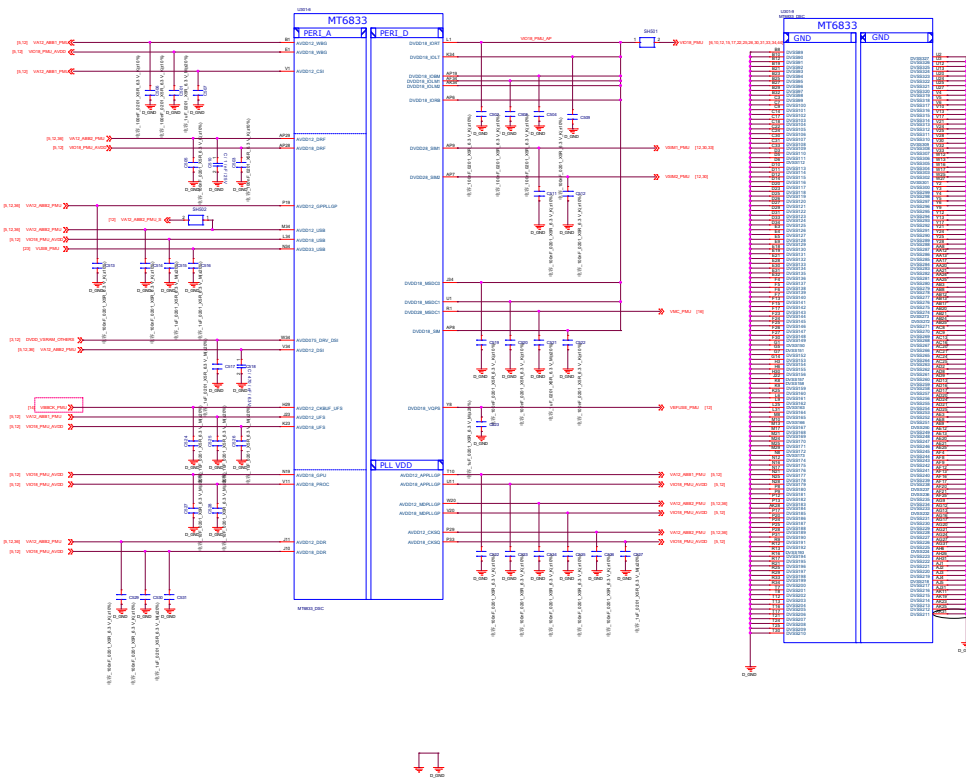


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 03.BB_POWER_PDN1			
Size D	Project <Doc>	Rev v1	
Date: Monday, October 18, 2021	Sheet 3 of 23		

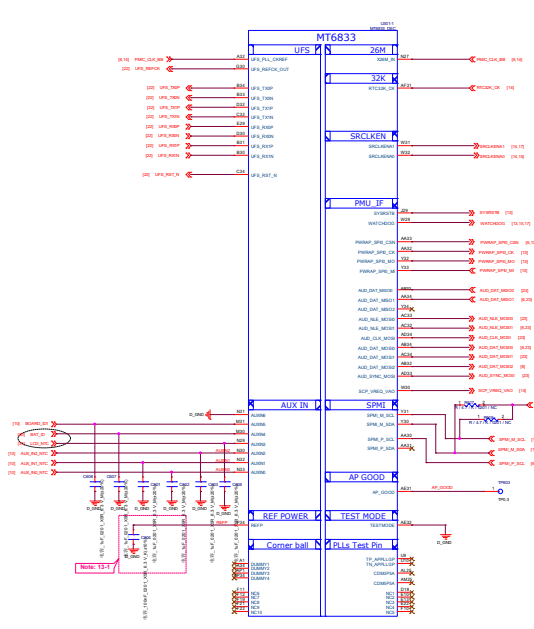


Schematic design notice of "11_BB_POWER_PDN" page.
 Note 11-1: Differential pair of PMC remote sense must be close to SIO's ball.
 Remote sense trace with GND shielding to PMC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 04.BB_POWER_PDN2			
Size D	Project <Doc>	Rev v1	
Date: Monday, October 18, 2021	Sheet 4 of 23		

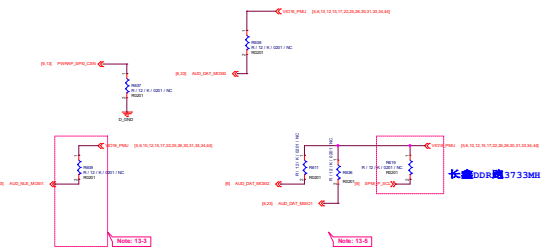


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		05.BB_POWER_IO	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet	5 of 23



Note 13-6: "AUD_DAT_MISO1" is trapping pin to select VEMC Voltage.

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V for VCC of UFS2.1 device
H (by external PU)	VEMC=2.5V for VCC of UFS3.0 device



Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MISO1" pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MISO1	AP_JTAG	ID_JTAG
H (Default)	L (Default)	N/A	N/A
(by external PU)	(by external PU)	SPINENT8	ENT21ENT22IDC7
L (by external PD)	L	SPI0-ENT8	MSDC1_CLK MSDC1_CMD MSDC1_DATA1 MSDC1_DATA2
H (by external PU)	H (by external PU)	MSDC1_CLK MSDC1_CMD MSDC1_DATA1 MSDC1_DATA2	N/A

Note 13-3: "AUD_NLE_MOSI1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOSI1	USB_JTAG
L (Default)	Disable serial JTAG over USB
H (by external PU)	DPIDM output JTAG

Note 13-4: "AUD_SYNC_MOSI" and "AUD_NLE_MOSI0" pin features in trapping pin to booting (eMMC/UFS/SPI NOR).

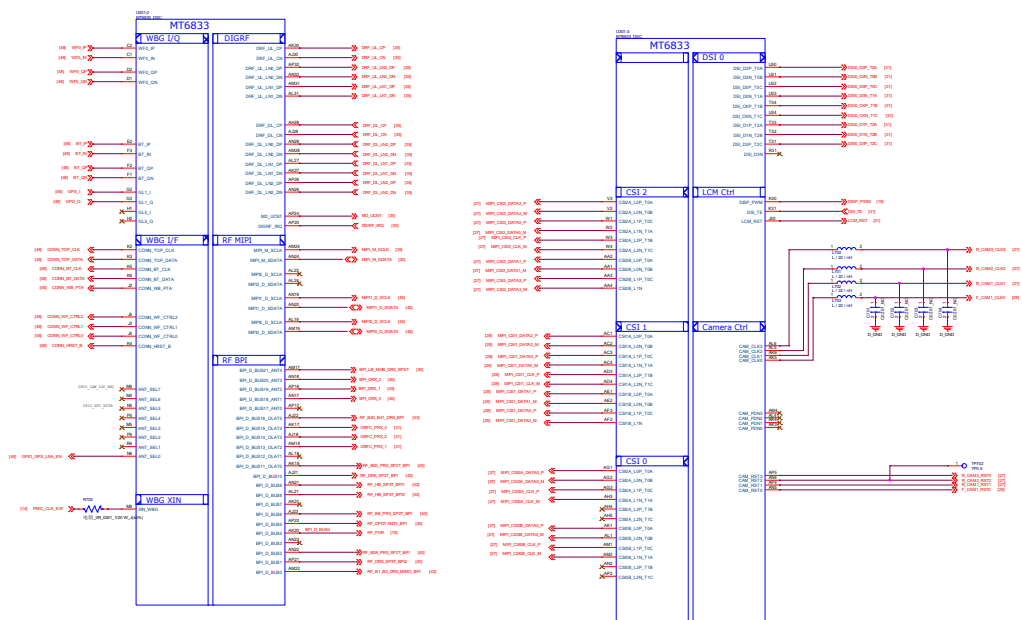
AUD_SYNC_MOSI	AUD_NLE_MOSI0	Storage Booting
L (Default)	L (Default)	Only UFS boot
H (by external PU)	H	Only eMMC boot
(by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved

Note 13-5: "AUD_DAT_MOSI1" and "AUD_DAT_MOSI2" pin features in trapping pin to enable DDR.

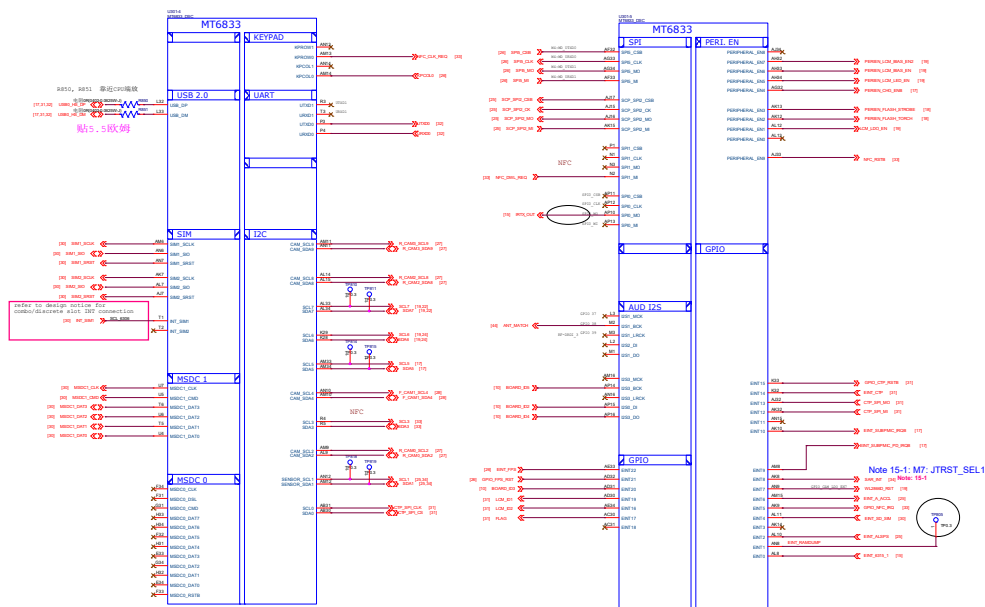
AUD_DAT_MOSI1	AUD_DAT_MOSI2	SPM_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4xMCP-4266A
H (by external PU)	H (by external PU)	L (Default)	LP4xDiscrete-4266A
(by external PU)	L	L (Default)	Reserved
H (by external PU)	H (by external PU)	L (Default)	Reserved

Note: 1. SPM: P_SCL for DDR 4266Mbps/3733Mbps Co-load, 1.2V = 1250Mbps.
2. (by external PU) = 3733Mbps

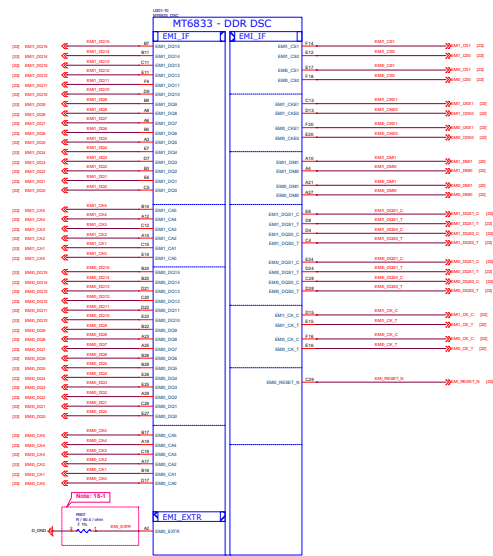
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 06.BB_1		
Size D	Project <Doc>	Rev v1
Date: Monday, October 18, 2021	Sheet 6 of 23	



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		07.BB_2	
Size	Project	<Doc>	Rev
D			v1
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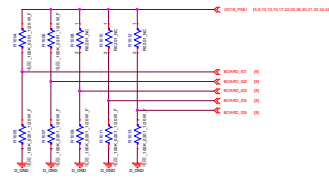
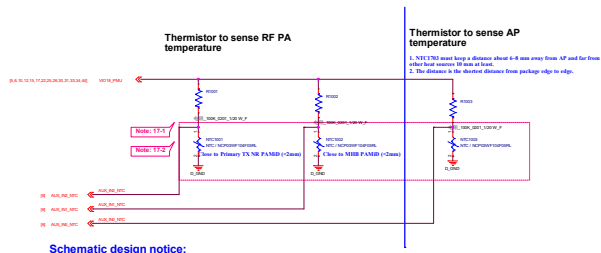
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
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Schematic design notice of "16_BB_4 Interface" page:

Note 16-1: R4001 please select 60.4 ohm (1%) resistor

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		09.BB_4	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet	9 of 23

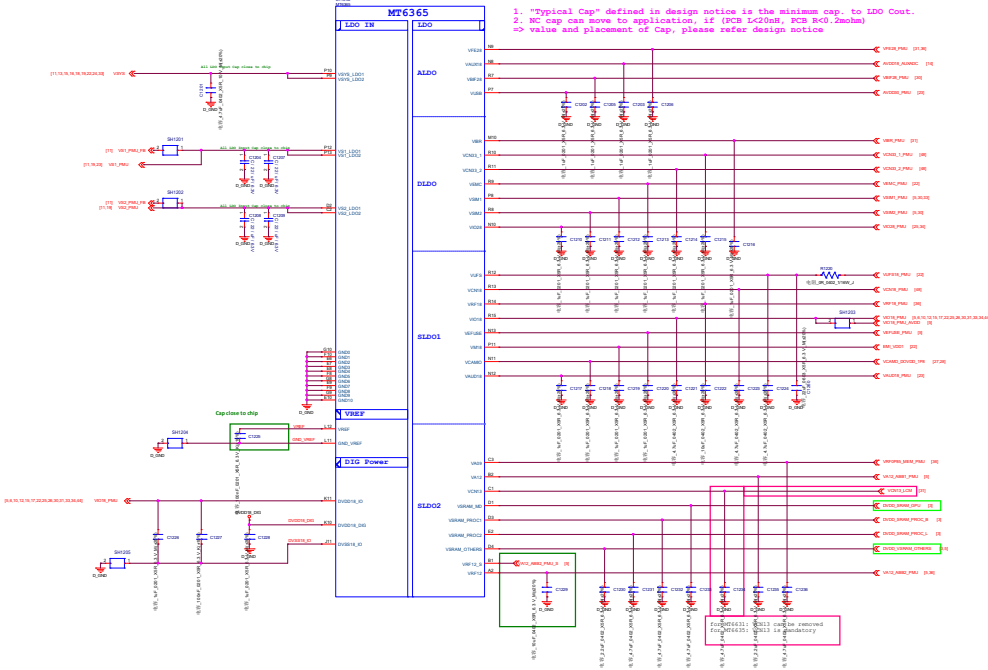


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		10.BB_AUXADC_Thermal/BOARD_ID	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet	10 of 23

Note: Refer to Platform SoC Design Notice for PMIC part number

MT6365

1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. MC cap can move to application, if (PCB 1x20m, PCB R40.2mohm) => value and placement of Cap, please refer design notice



Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
ALDO	VFE2B	2.80	OFF (2.8)	200	RFEE
	VU1B	3.07	OFF (3.07)	200	U1B
	VALUX1B	1.84	OFF (1.84)	50	ALUXADC
	VIO22	2.24	ON (2.24)	25	DCKO
VRTC	VRTIC	2.80	OFF (2.8)	50	Battery Interface
	DIVDD1B_D10	1.80	ON (1.8)	10	PMIC Digital

Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
BLDO	VSM1	1.7/1.8/1.9/2.0	OFF (1.8)	200	SIM
	VSM2	2.7/3.0/3.3	OFF (3.0)	300	SIM
	VCM33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCM33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.7/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VIO2B	2.8/2.9/3.0	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.0/3.3	OFF (2.8)	200	Vibrator

Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
SLDO1	VU1B	1.80	ON (1.8)	200	Audio
	VCM40	1.80	OFF (1.8)	200	Camera IO
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor

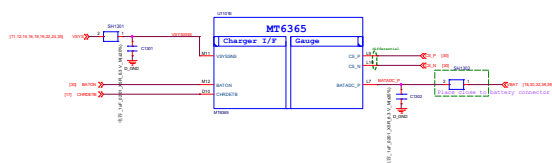
Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
SLDO2	VU1B	1.80	ON (1.8)	200	Audio
	VCM40	1.80	OFF (1.8)	200	Camera IO
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor
	VU1B	1.80	ON (1.8)	200	IO & Sensor

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 12.POWER_MT6365_LDO

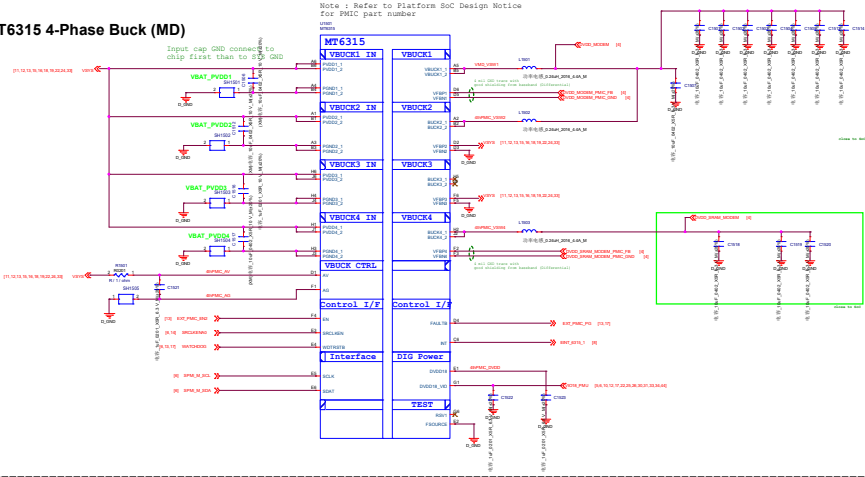
Size D Project <Doc> Rev v1

Date: Monday, October 18, 2021 Sheet 12 of 23



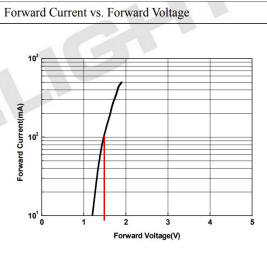
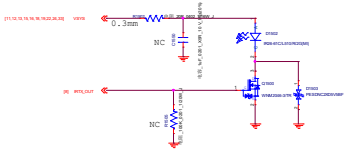
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 13.POWER_MT6365_General			
Size D	Project <Doc>	Rev V1	
Date: Thursday, November 11, 2021			
		Sheet 13 of 23	

MT6315 4-Phase Buck (MD)



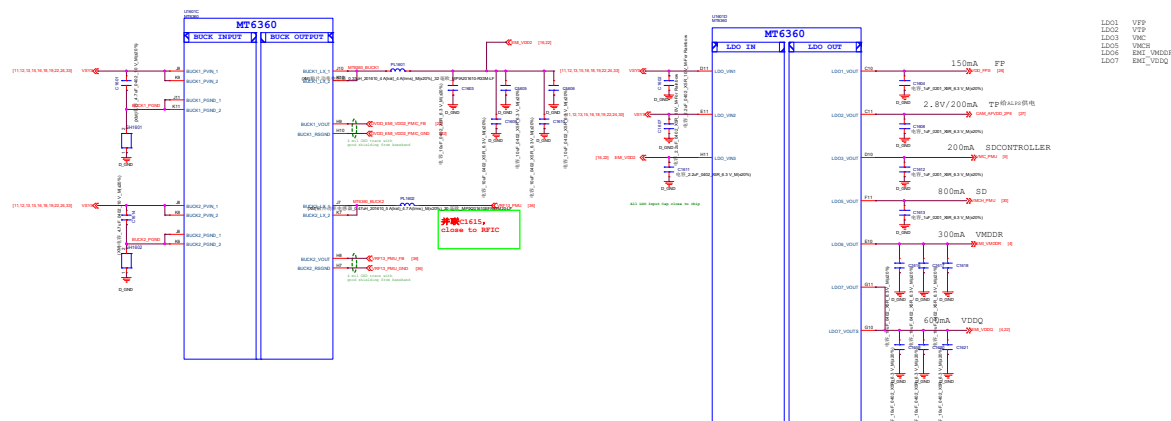
MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAx} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	---
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 15.POWER_MT6315_VMD/IR_LED			
Size D	Project <Doc>	Rev v1	
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Note : Refer to Platform SoC Design Notice for PMIC part number

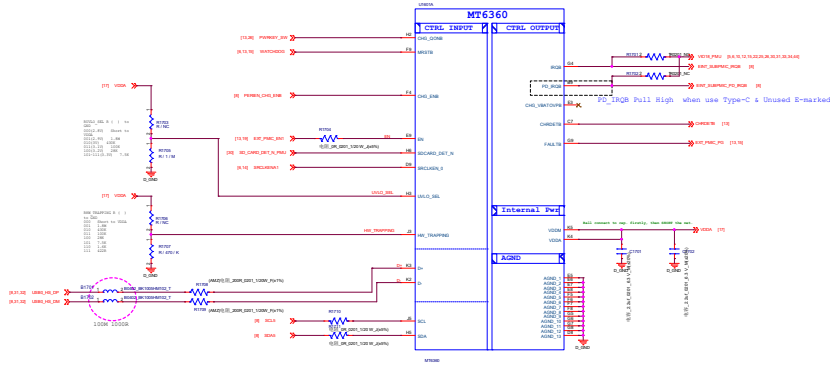


Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

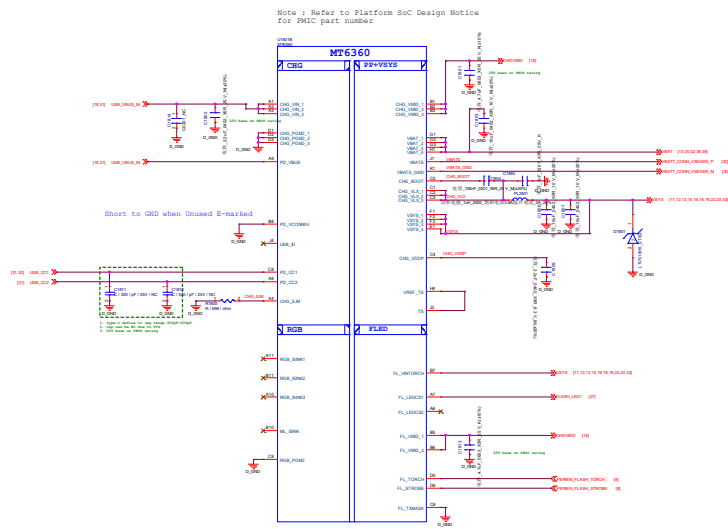
Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 16.POWER_MT6360_BUCK_LDO*		
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Note 1: Refer to Platform SoC Design Notice for PMIC part number



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Title 17.POWER_MT6360_General			
Size	Project <Doc>		Rev
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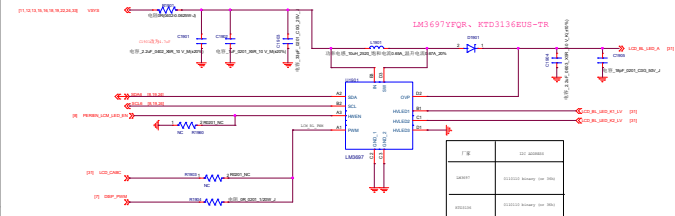


Schematic design notice of "28_POWER_MT6360_Charger" page.

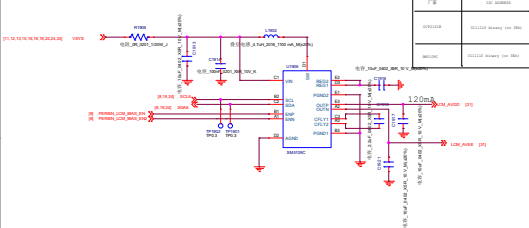
Note 25-1: For better ESD & surge performance we need choose suitable device for system protection. Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		18.POWER_MT6360_Charger	
Size	Project	<Doc>	Rev
D			V1
Date:	Friday, December 24, 2021	Sheet 18 of	23

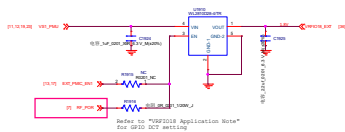
LCM Backlight LED Driver



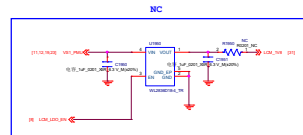
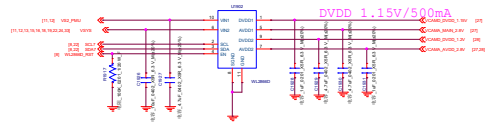
LCM Bias



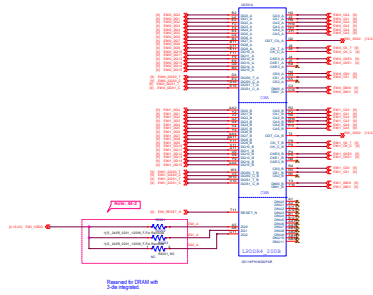
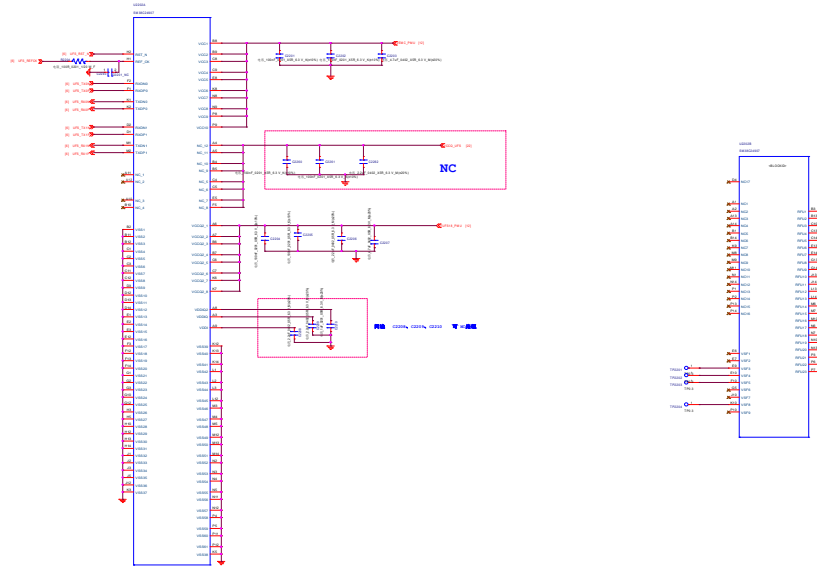
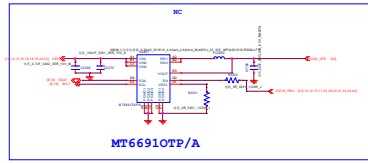
External VIO18_LDO: option for power leakage from UART port on product line



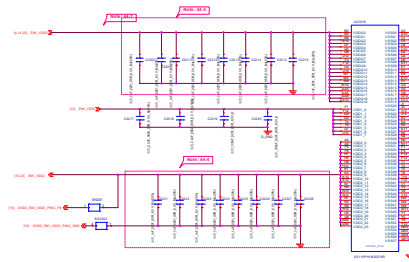
LED Array	LED Array
12mA	1.5V
12mA	1.5V



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 19.POWER_EXT		
Size D	Project <Doc>	Rev v1
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discrete LP4X
16Gb LPDDR4X



Schematic design notice of "44_Memory_4MMC_LPDDR4X"

Note 44-1: Please refer to power supply related page select: VDRAM2 / VDRAM output voltage property for LPDDR4X.

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

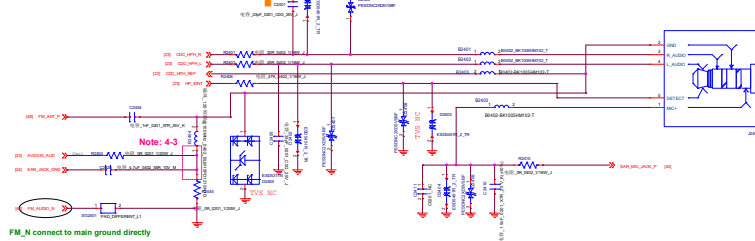
Note 44-3: Please refer to eMMC vendor's datasheet or MTK common design notice to get the recommendation typical cap. value for VDDQ/VDDQ2/VDDQ3 power domains of eMMC.

Note 44-4: VDDQ VDDQ2 decoupling cap. (closed to DRAM bank). For other cap for PMIC >10uF, at PMIC page, please also refer to PMIC and layout guide for placement.

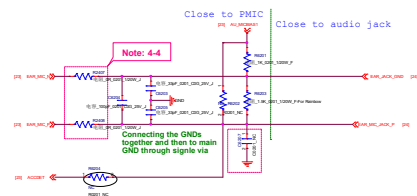
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 01_Block_Diagram		
Size E	Project J19_MB_V1	Rev v1
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Earphone Microphone

D2406, D2407, D2408 (MC), D2409
D2401, D2402, D2403 (MC), D2404



Earphone Microphone



Note 4-3:

Signal	Value	Unit
EAR_MIC	10K	Ω
EAR_SPK	10K	Ω

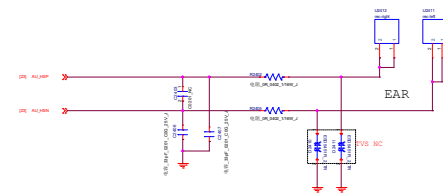
Note 4-4:

1. Need do SW configuration for DCC mode

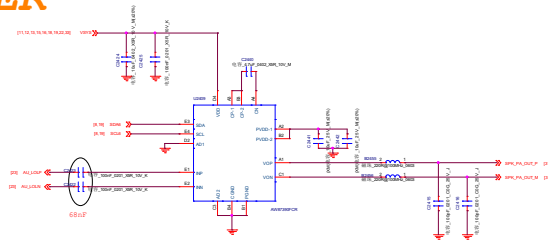
2. Analog input impedance is as below, please customer check the detail requirement with mic vendor if the mic used has special requirement on the input impedance of analog input

REF337	UL Mode	Rin	Typ	Max	Unit
C	DCC	-	-	51p	F

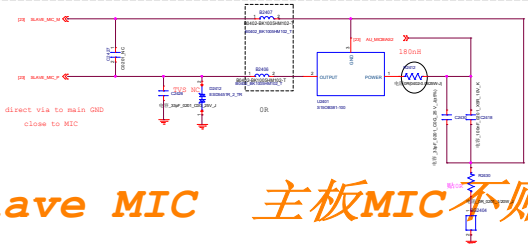
RECIVER



SPEAKER



Slave MIC 主板MIC不贴



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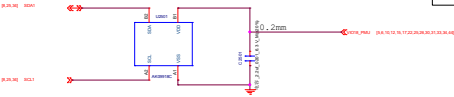
Title 24.MIC/RCV/SPK/Earphone

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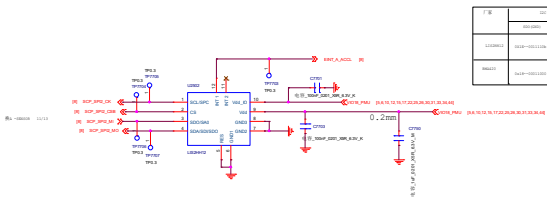
M-Sensor

厂号	120 400000
物料描述	0ch
物料描述	2ch

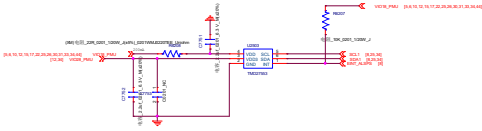


A sensor

default SPI (if sensor hub support)



ALS+PS

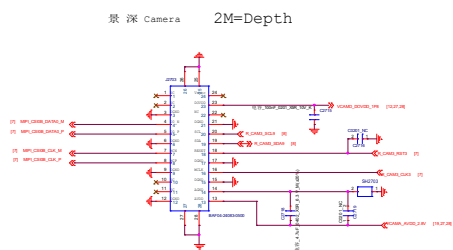
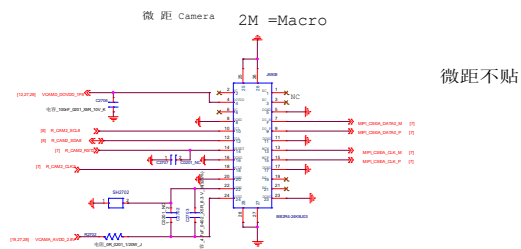
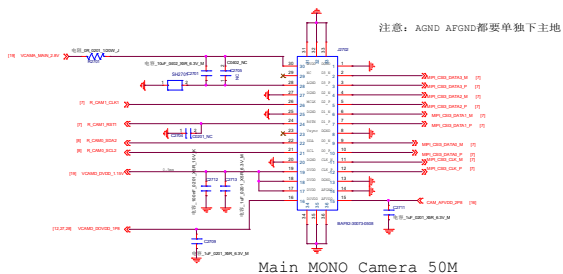


华勤通讯 Huaqin Telecom Technology Com.,Ltd

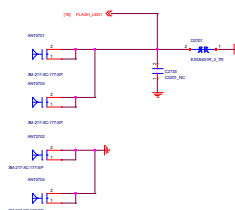
Title 25.SENSOR

Size D	Project <Doc>	Rev V1
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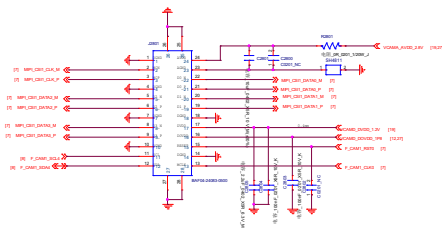
Flash LED 5V Boost



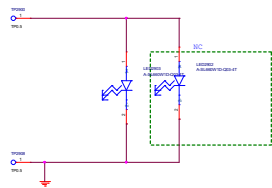
Flash 下方地要完整且足够大, 散热一定要好

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		27.REAR_CAM	
Size	Project	<Doc>	Rev
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Front Camera - 8M

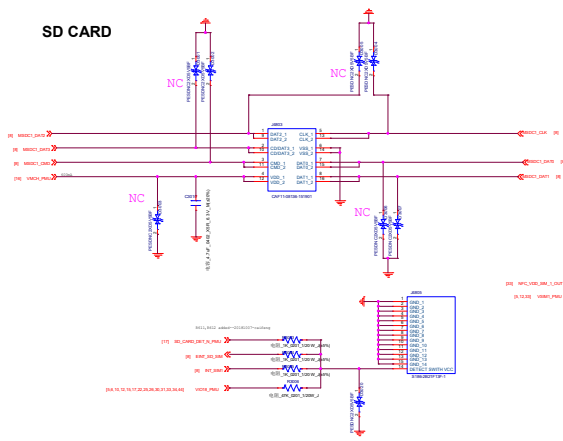


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		28.FRONT_CAM	
Size	Project	<Doc>	Rev
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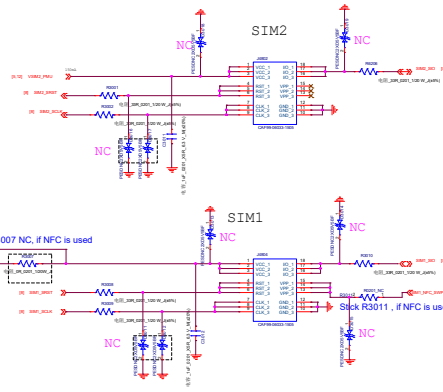


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		Sub_Charger_SMB1396	
Size	Project	<Doc>	Rev
D			V1
Date: Monday, October 18, 2021		Sheet 29 of	50

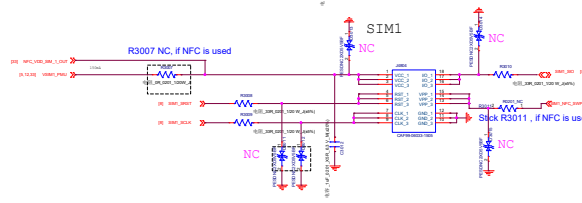
SD CARD



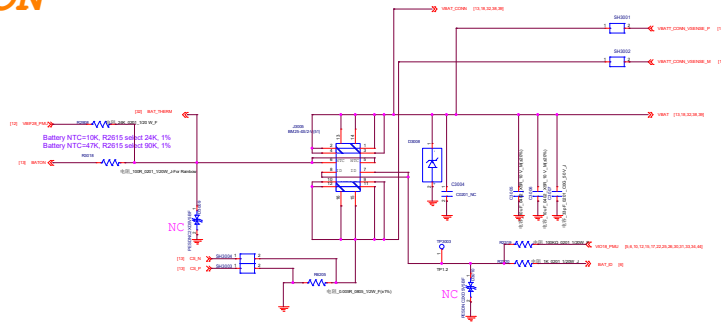
SIM2



SIM1

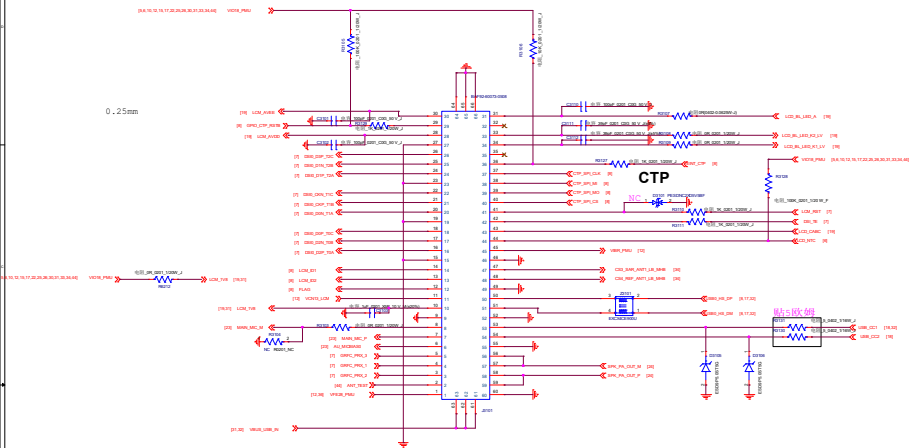


BATT CON

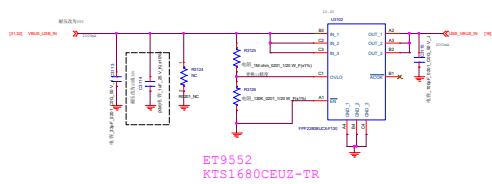


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		30.SIM/BATT	
Size	Project	<Doc>	Rev
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LINK



OVP



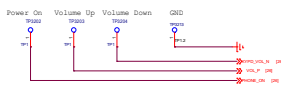
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		31.60PIN_LCD_LINK/OVP	
Size	Project	<Doc>	Rev
D			V1
Date: Friday, December 24, 2021		Sheet 31 of 50	

VBAT TEST POINT

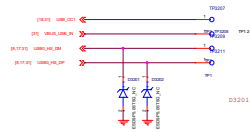


VBAT

KEY BTB

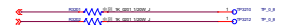


KEYPAD

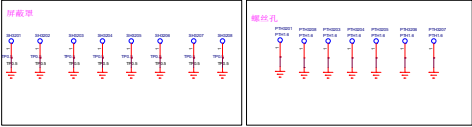


VCHG

UART TP

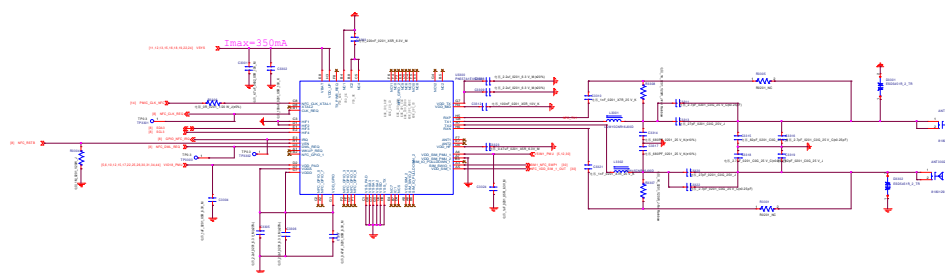


JTAG



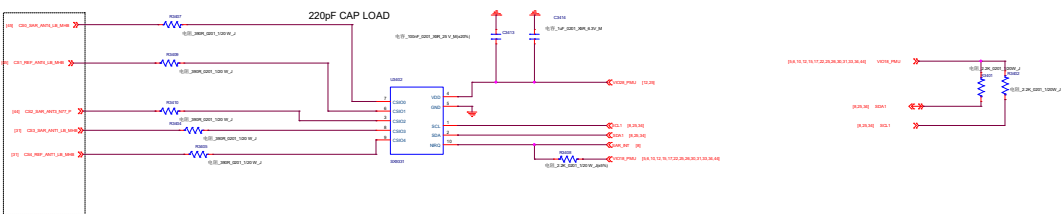
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev V1
Date: Monday, October 18, 2021	Sheet 32 of 50	

RF_NFC



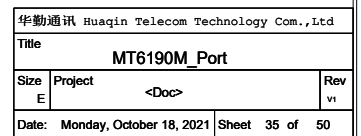
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title RF_NFC			
Size E	Project Doc	Rev vi	
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SAR_sensor



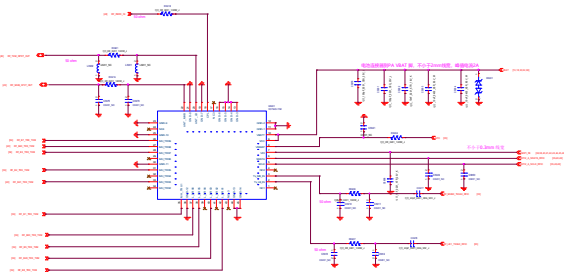
Design notice:
1) CS0_1 nets are differential pairs;
2) The width for each line, should be less than 0.1mm;
3) 0.2mm should be guaranteed between the ground and differential pairs;
4) And also, the distance between the differential line should be more than 0.2mm;
5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title			
SAR_sensor			
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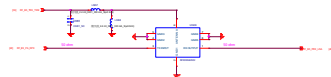
5
MAIN_ANT TXM_GSM

LB
MHB

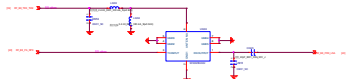


TRX_LB

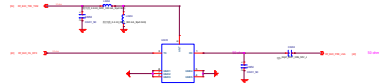
B5 TX



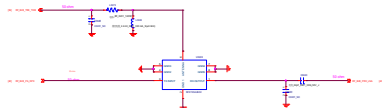
B8 TX



B20 TX

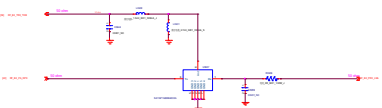


B28 TX

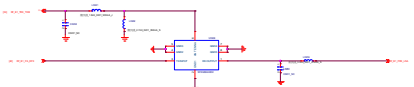


TRX_MHB

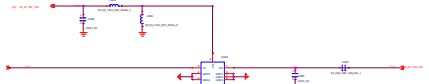
B2 TX



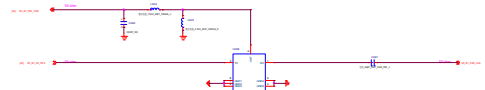
B1 TX



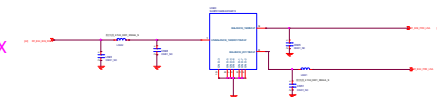
B3 TX



B7 TX

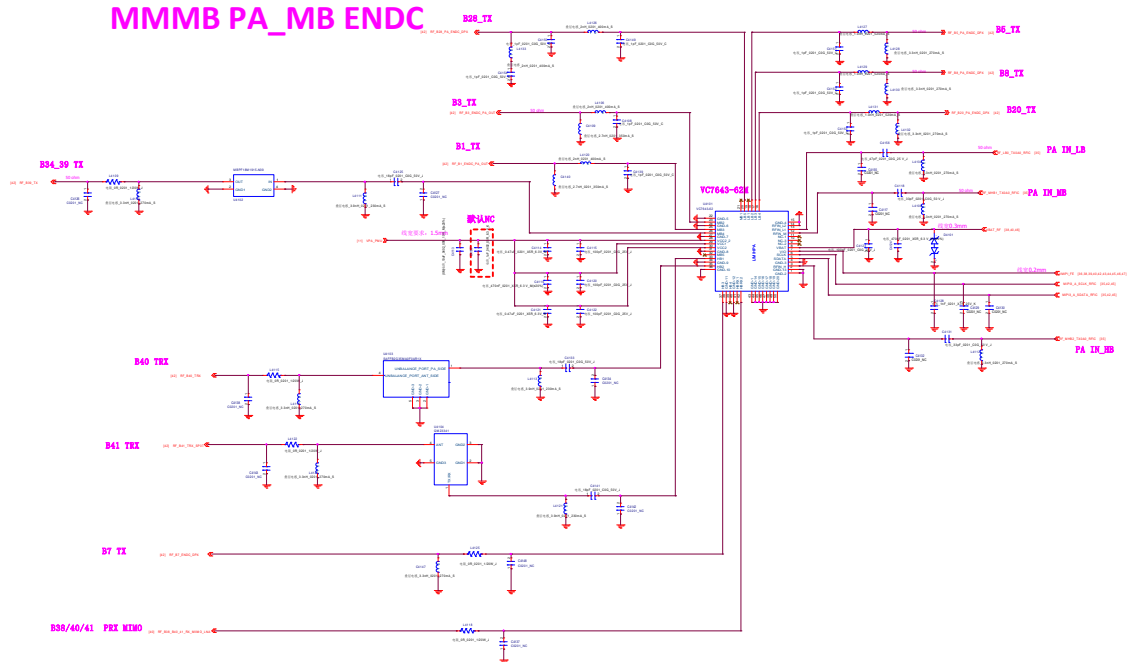


B34 /B39 PRX



华勤通讯 Huaqin Telecom Technology Co., Ltd			
Title TX_Module			
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Custom			v1
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MMMB PA_MB ENDC



华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title MMBB PA_MB ENDC		
Size E	Project <Doc>	Rev v1
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D

D

C

C

MHB

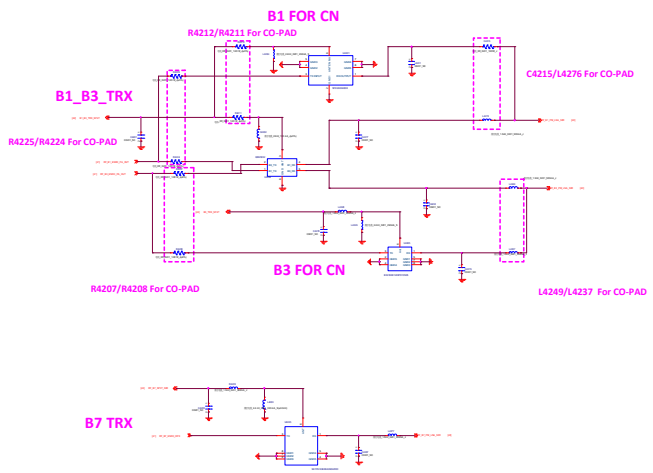
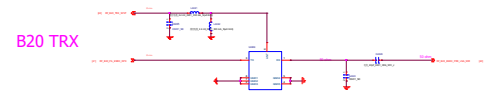
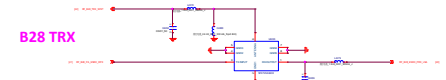
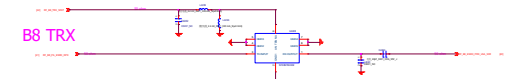
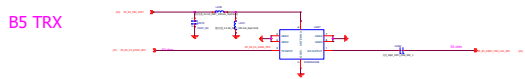
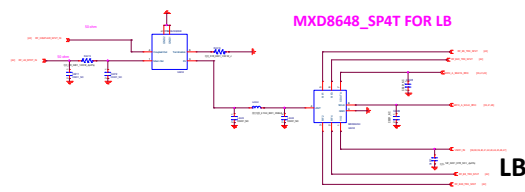
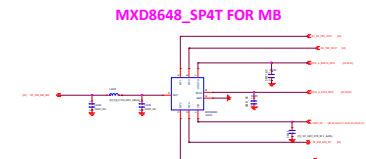
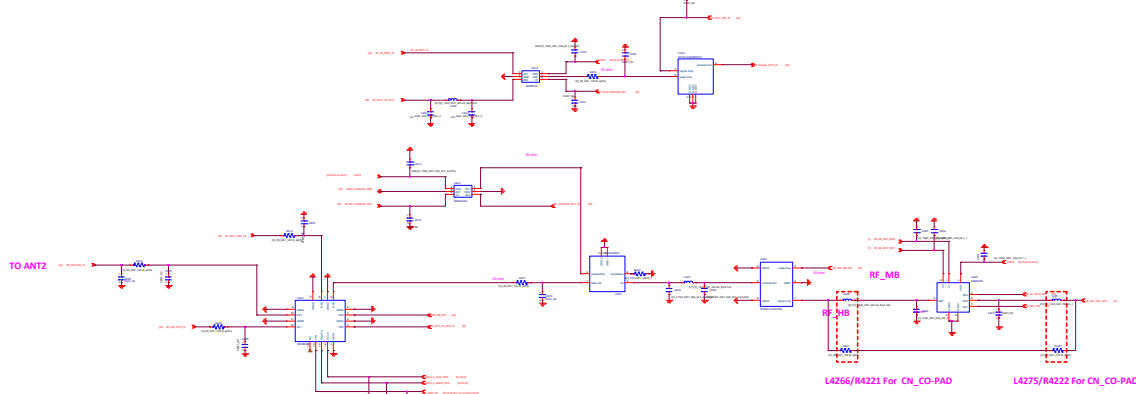
C

B

B

A

A



华为通信 Huawei Telecom Technology Co., Ltd			
Title: MMMB PA_VC7643_ENDC_MHB_TRX			
Size	Project	<Doc>	Rev
Custom			v1
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PIN TO SFHG40AA002-B1 RX SAW

B1 DRX MIMO For CH

L4368/L4369 FOR CO-PAD

B1 DRX MIMO For POCD

B3 DRX MIMO For POCD

B7 DRX MIMO For POCD

B40 DRX MIMO For POCD

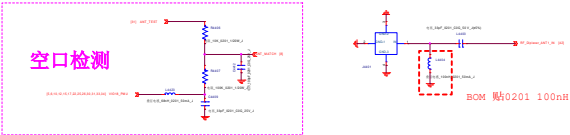
SP2T_MXD8625C

B41 DRX MIMO

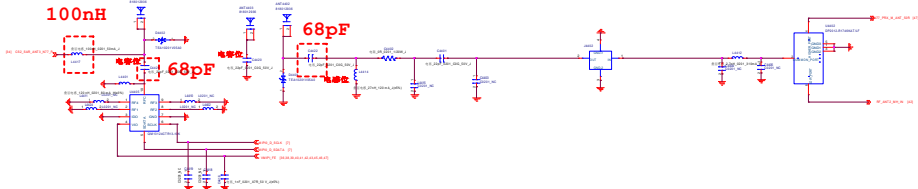
To ANT5

华勤通讯 Huaqin Telecom Technology Co.,Ltd			
Title MXD98E2DA_VC3725			
Size	Project		Rev
Custom	<Doc>		v1
Date: Friday, October 22, 2021		Sheet 43 of 50	

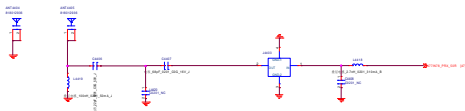
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_PRX_MIMO)

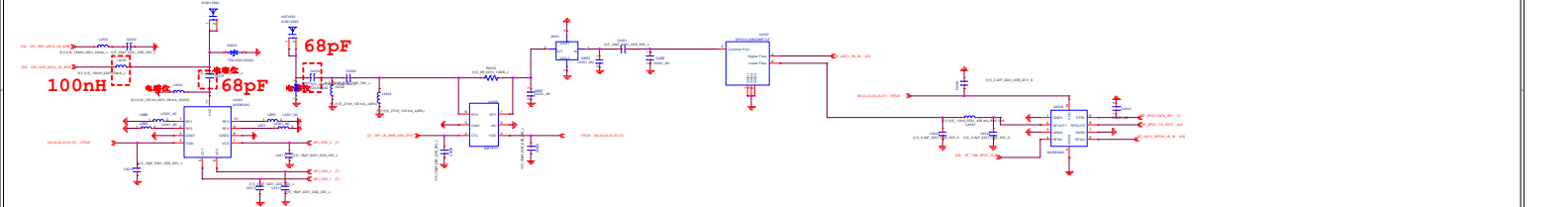


ANT3 (N77_PRX)

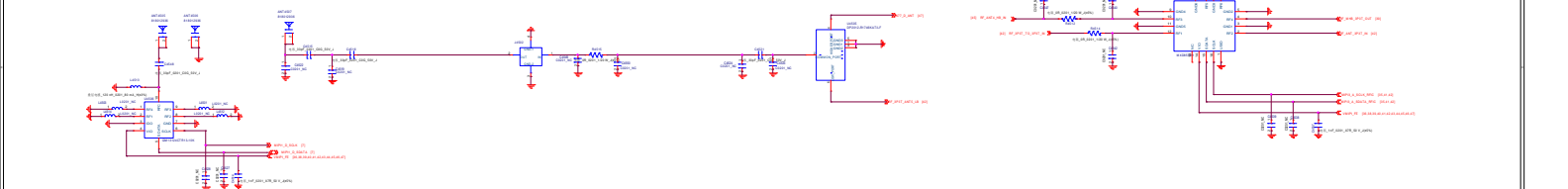


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		ANT_P_PM	
Size	Project	<Doc>	Rev
E			v1
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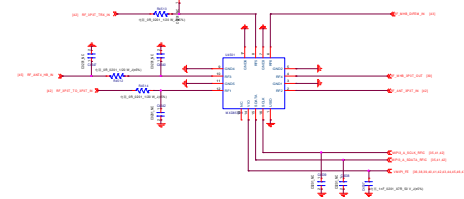
ANT4 (LB_MHB_D)



ANT5 (MHB_DM/N77_D)



AsDIV_3P3T



接地弹片



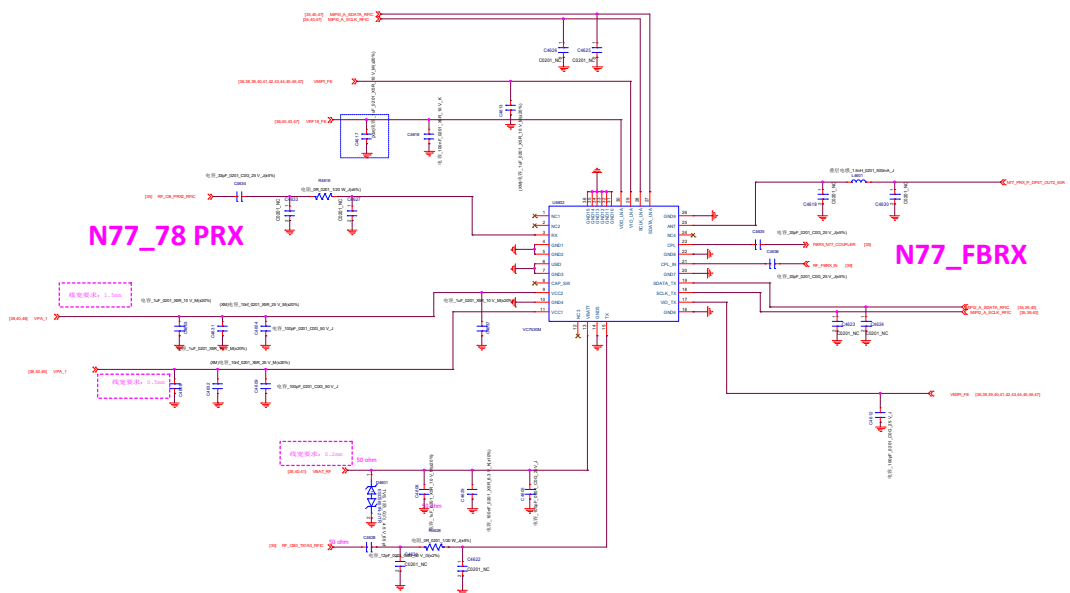
ANT6 (N77_DM_HB_DRX) to 小板



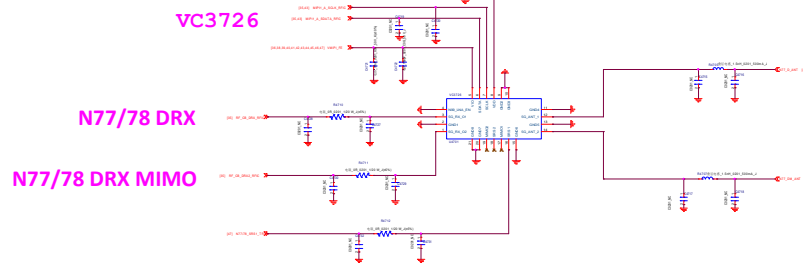
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		ANT_D_DM	
Size	Project	<Doc>	Rev
E			v1
Date:	Friday, October 22, 2021	Sheet 45 of	50

N77/78_TRX0_VC7530M

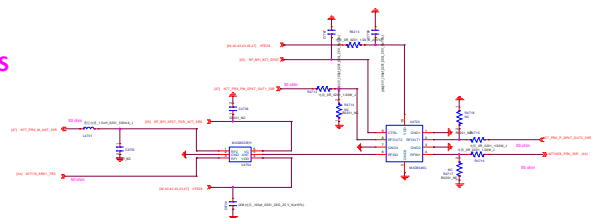
NTC温度检测
见P10的R1002 and NTC1002



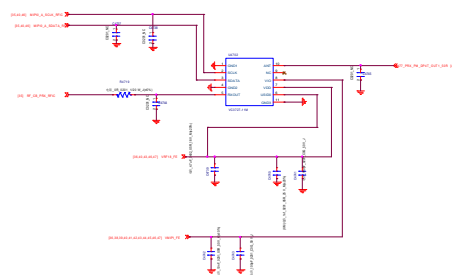
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		N77/78_TRX0_VC7530M	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet	46 of 50



DPDT+SP2T FOR N77/78 SRS

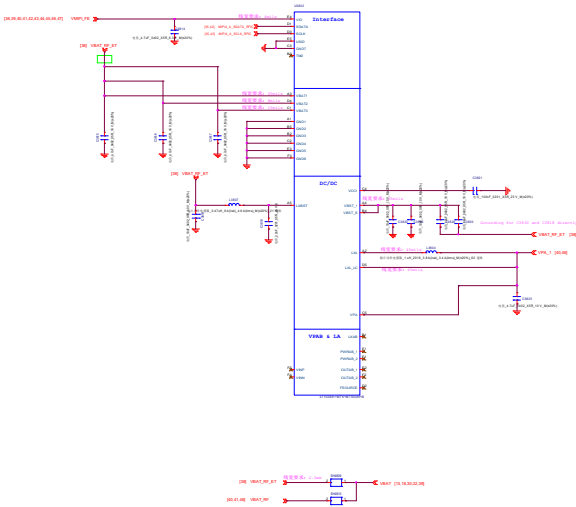


VC3727-11M_N77/78 PRX MIMO



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		VC3726	
Size	Project	<Doc>	Rev
E			v1
Date: Saturday, November 13, 2021		Sheet 47 of 50	

MT6308 FOR Phase 5N PA_N77_PA



华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		RF_APT_MT6308M	
Size	Project	<Doc>	Rev
E			v1
Date: Monday, October 18, 2021		Sheet 50 of 50	